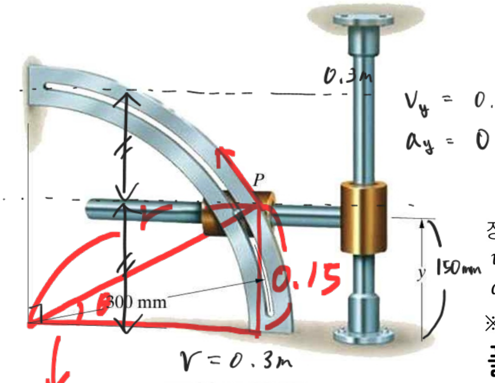
13.89 If $y = 150$ mm, $dy/dt = 300$ mm/s, and $d^2y/dt^2 = 0$, what are the magnitudes of the velocity and acceleration of point P?



Problem 13.89

$V_y = 0.3\,m/s$
$a_y = 0$
등속 운동

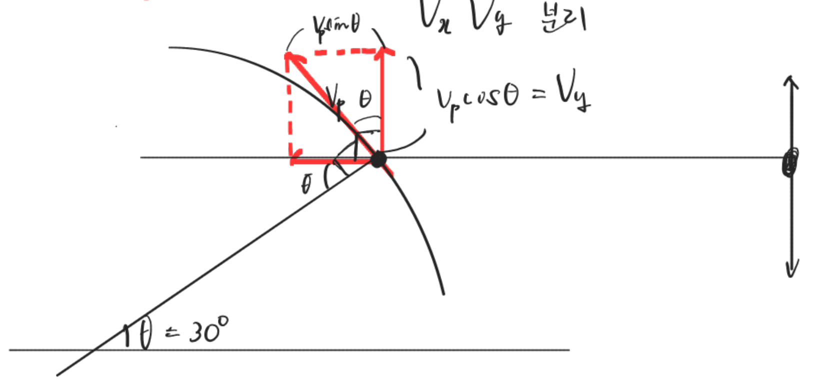
정답
$v_p = 0.3464\ m/s$
$a_p = 0.4619\ m/s^2$
※마찰력 등 모든 외력 무시

풀이과정 자세히 서술

$\sin\theta = \frac{0.3}{0.15} = \frac{1}{2}$

$\therefore \theta = 30°$

$V_x\ V_y$ 분리



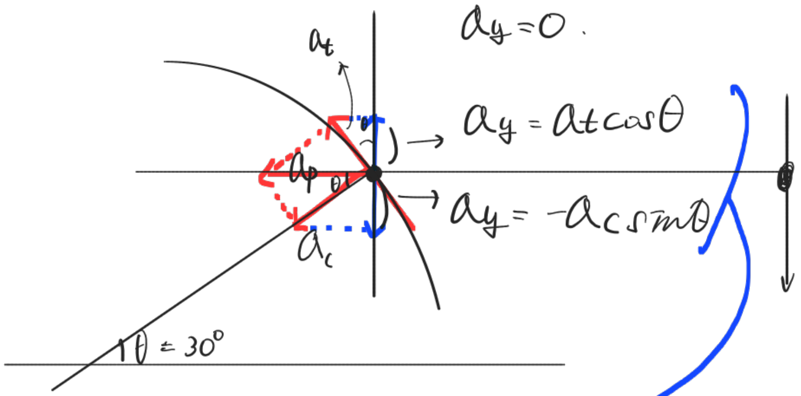
$V_p\cos\theta = V_y$

$V_y = 0.3\,m/s$

$$\therefore V_y = V_p\cos 30° = 0.3\ m/s$$

$$V_p = \frac{0.3}{\cos 30°}\ m/s$$

$$= \frac{0.6}{\sqrt{3}} \approx 0.3464\ m/s$$



$a_y = 0$.

$a_y = a_t\cos\theta$

$a_y = -a_c\sin\theta$

구심 가속도 공식

$$a_c = \frac{V^2}{r} = \frac{(0.3464)^2}{0.3}$$

$$a_t \cos 30^\circ - a_c \sin 30^\circ = 0 .$$

$$\therefore \quad a_t \cos 30^\circ = a_c \sin 30^\circ$$

$$a_t = \frac{(0.3464)^2}{0.3} \cdot \frac{\sin 30^\circ}{\cos 30^\circ}$$

$$a_p = \sqrt{a_t^2 + a_c^2}$$

$$= \sqrt{\frac{(0.3464)^4}{(0.3)^2} \cdot \frac{(\sin 30^\circ)^2}{(\cos 30^\circ)^2} + \frac{(0.3464)^4}{(0.3)^2}}$$

$$= \frac{(0.3464)^2}{0.3} \sqrt{\left(\frac{(\sin 30^\circ)^2}{(\cos 30^\circ)^2} + 1 \right)}$$

$$\approx 0.4619 \, m/s^2$$